Modelling with Differential Equations

Exam code: 9MA0

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Written by: Paul
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Help me understand

Modelling with differential equations

What can be modelled with differential equations?

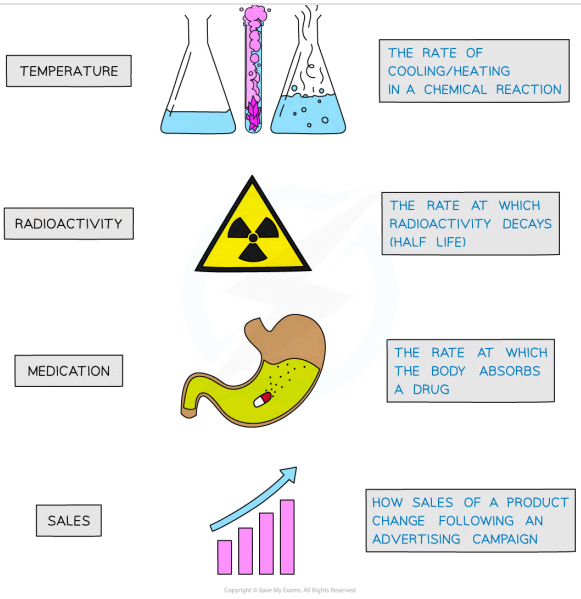
- Derivative terms like $\frac{dy}{dx}$ are "rates of change"
- There are many situations that involve "change"

- Temperature
- Radioactivity
- Medication
- Sales

Stuck on something?

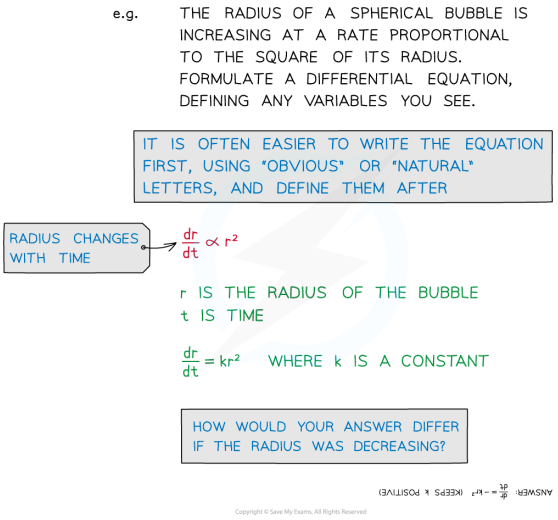
Ask questions and get help with exactly what you're revising.

Got it



How do I set up a model with differential equations?

- The first task is to set up a differential equation from a description in words:



- Important phrases here are …
  - … "**rate of change**" … reference to a derivative term like $\frac{dy}{dx}$
  - … **directly/inversely proportional** to … $y = kx$, $y = \frac{k}{x}$
  - … **formulate** … means to write as an equation
    - you may need to choose and define letters for variables
    - **V** for volume, **h** for height (of a cylinder, say)

TEMPERATURE OF ITS SURROUNDINGS (AMBIENT TEMPERATURE).
USING T AS THE TEMPERATURE OF A CUP OF TEA, t FOR TIME AND T_0 AS THE TEMPERATURE AROUND THE CUP WRITE DOWN A DIFFERENTIAL EQUATION FOR NEWTON'S LAW OF COOLING.

- "RATE OF" – DEPENDENT ON TIME t
- PROPORTIONAL TO... $\propto$
- T IS "CURRENT" TEMPERATURE DIFFERENCE BETWEEN T AND T_0

YOU MAY HAVE TO READ THE QUESTION SEVERAL TIMES.

$$\frac{dT}{dt} \propto T - T_0$$

IF COOLING, $T_0 < T$

$$\frac{dT}{dt} = -k(T - T_0) \quad \text{WHERE } k \text{ IS A CONSTANT}$$

COOLING SO EXPECTING A NEGATIVE IMPLIES $k>0$

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- Some differential equations may involve Connected Rates of Change

e.g. A RIGHT–ANGLED TRIANGLE HAS A BASE OF **3p** cm AND A HEIGHT OF **2p** cm, WHERE **p > 0** IS A PARAMETER THAT IS INCREASING OVER TIME. THE RATE OF INCREASE OF **p** IS SUCH THAT THE RATE OF INCREASE OF THE THE AREA OF THE TRIANGLE, **A**, IS PROPORTIONAL TO THE SQUARE ROOT OF **p**. WRITE DOWN A DIFFERENTIAL EQUATION FOR THE RATE OF CHANGE OF **p**.

$$\frac{dA}{dt} \propto \sqrt{p} \quad \text{SO} \quad \frac{dA}{dt} = k\sqrt{p}$$

A IS INCREASING SO $k > 0$

WHERE k IS A POSTITVE CONSTANT

WE KNOW THIS

BY THE CHAIN RULE, $\frac{dA}{dt} = \frac{dA}{dp} \times \frac{dp}{dt}$

USE THE CHAIN RULE TO CONNECT WHAT YOU KNOW AND WHAT YOU WANT TO KNOW

WE WANT TO KNOW THIS

THE AREA OF THE TRIANGLE IS

$$A = \frac{1}{2} \times 3p \times 2p = 3p^2$$

USE DIFFERENTIATION TO GET THE 'MISSING' DERIVATIVE FROM THE CHAIN RULE EQUATION

$$\text{SO} \quad \frac{dA}{dp} = 6p$$

$$\text{THEN} \quad \frac{dA}{dt} = \frac{dA}{dp} \times \frac{dp}{dt}$$

$$k\sqrt{p} = 6p\left(\frac{dp}{dt}\right)$$

SUBSTITUTE AND REARRANGE

$$\text{AND} \quad \frac{dp}{dt} = \frac{k\sqrt{p}}{6p}$$

$$\frac{dp}{dt} = \frac{k}{6\sqrt{p}} \quad \text{(WHERE } k \text{ IS A POSITIVE CONSTANT)}$$

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Examiner Tips and Tricks

- Use a highlighter (or underline) to pick out important words/phrases
- Read and re-read the question several times

Worked Example

Teacher only

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